

Variable Gain Linear Quadratic Regulator and Its Application*

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Abstract—A variable gain linear quadratic regulator(VGLQR) for stabilizing a class of multivariable nonlinear control system is proposed. The VGLQR method is derived by using the Lyapunov function approach. This control scheme can be viewed as a novel gain scheduling controller, and the feedback gain obtained by solving algebraic Riccati equation online is changing with state. In addition, a fast algorithm based on Schur method for the solution of algebraic Riccati equations online is discussed and a software package was developed during the present research. In this paper, the VGLQR control method is applied to a general nonlinear system, also the effectiveness of the proposed method is verified by stabilization of triple inverted pendulum system. Simulation and physical experiment results demonstrate that the VGLQR is of good stability and adaptability, which are distinctly better than that of classical LQR controller.

Index Terms—Variable gain linear quadratic regulator(VGLQR), Algebraic Riccati equation (ARE), Triple inverted pendulum system.

I. INTRODUCTION

The optimal linear quadratic regulator (LQR) theory [1], [2] are well-established in 1960s and is widely applied in linear systems. On the other hand, the theories of nonlinear regulation emerged during the late 20th century, and are always the research highlights. Recently, many theories and methodologies appeared for the control of highly nonlinear systems have been widely applied in handling systems with various structures or properties, such as the feedback linearization method [3], gain scheduling [4]–[6], dynamic inversion [7]–[10], sliding mode control [11], [12], recursive backstepping [13], [14], state-dependent Riccati equation(SDRE) technique [15], [16], and variable universe adaptive fuzzy control method [17], [18], etc. For nonlinear control theory, the control design methods must be focused on performance such as stability, optimality, robustness and adaptability. Many of the nonlinear regulation methods used

today are approximate methods which attempt to exploit linear system theory. The feedback linearization method [3] and the gain scheduling control methods [4]–[6], [15] all share as a common core the use of linear controller design techniques to address nonlinear problems. The effectiveness of recursive backstepping [14] has been demonstrated in systems having the cascaded structure, and the recursive interlacing design method [19] was developed for designing globally stabilizing robust control for general systems. Inversion of system dynamics [7]–[10], [20] can be used to find feedforward inputs that achieve high precision positioning.

The control method that is proposed in this paper, the variable gain linear quadratic regulator(VGLQR) method, aims to provide a way of stabilizing general time-invariant nonlinear systems. The VGLQR method is derived by using the Lyapunov function approach, and the linear quadratic regulation (LQR) theory is employed to address nonlinear regulation problem. The significant feature of the VGLQR is that feedback gain is changing with state variables, which is assumed to be measured. In practice all measurements are only taken at discrete sampling instants, and the control action must be piecewise constant, and actually the VGLQR control method is equivalent to using LQR method to obtain optimal feedback gain in every sampling interval. Therefore, the VGLQR method can guarantee asymptotic stability of the nonlinear control system, and to some extent has the desirable suboptimality property.

The VGLQR method requires the computation of the Jacobi matrices of the nonlinear system and the solution of algebraic Riccati equation online in order to obtain nonlinear feedback control. However, it is difficult to find the analytic solution for algebraic Riccati equation, especially high-order systems. Among the numerical techniques for solving algebraic Riccati equation, two famous algorithms, the Kleinman iterative algorithm [21] and the Schur method [22], are used extensively. The Schur algorithm is apparently more numerically stable and performs reliably on systems

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with dense matrixes of up to order 100 or so. In view of this, the Schur algorithm is discussed in detail in the paper, and a software package, which is used to solve algebraic Riccati equation online, is developed during the present research. The experiments demonstrate that the online algorithm based on Schur algorithm, designed in this paper, has a high speed of operation, overcomes the difficulties in engineering application and can finish the decomposition of a large scale matrix in a short time.

The VGLQR method has been successfully implemented in stabilizing mult-rod inverted pendulum, and the good stability and adaptability is displayed. This paper aims to discuss the VGLQR method and its applications in detail. The paper is organized as follows. In Section 2, we will briefly review the optimal LQR theory and the VGLQR method is discussed in detail. In Section 3, two methods for solving algebraic Riccati equation are analyzed, and a algorithm for online solving algebraic Riccati equation is developed. In Section 4, the VGLQR method is applied to stabilize a triple inverted pendulum. Both the simulation results and the physical experiment results are displayed in this section. Finally in Section 5, some concluding remarks are made.

II. THE VGLQR METHOD

The optimal LQR problem is to find a control that makes the following criteria as small as possible

$$\min_u J = \frac{1}{2} \int_0^\infty (\mathbf{y}^T Q \mathbf{y} + \mathbf{u}^T R \mathbf{u}) dt, \quad (1)$$

where Q is an $l \times l$ symmetric positive definite matrix, R an $m \times m$ symmetric positive definite matrix. Solution to the optimal state-feedback LQR Problem. The optimal state-feedback LQR controller for the criteria(1) is a simple form as $\mathbf{u} = -K\mathbf{x}$, where K is the $m \times n$ matrix given by $K = R^{-1}B^T P$, and P is the unique positive definite solution to the following algebraic Riccati equation

$$PA + A^T P - PBR^{-1}B^T P + C^T Q C = 0. \quad (2)$$

Before proceeding, we shall summarize some well known results on (2) pertinent to our development. Kalman, the originator of the problem, has proved the following condition in [1].

Lemma 1 If (A, B) is completely controllable, then there exists a solution $P \geq 0$ to (2).

In [2] the following fundamental sufficient condition is proved by Kalman.

Lemma 2 If (A, B) is complete controllable and (C, A) is completely observable, then there exists a unique solution $P > 0$ to (2).

The two conditions are useful and fundamental for designing VGLQR controller.

Consider the autonomous nonlinear system

$$\begin{aligned} \dot{\mathbf{x}} &= \mathbf{f}(\mathbf{x}, \mathbf{u}), \quad \mathbf{x}(0) = \mathbf{x}_0, \\ \mathbf{y} &= \mathbf{h}(\mathbf{x}), \end{aligned} \quad (3)$$

where $\mathbf{x} \in \mathbb{R}^n$ is the state, $\mathbf{u} \in \mathbb{R}^m$ is unconstrained control input, $\mathbf{y} \in \mathbb{R}^l$ is the measured output. $\mathbf{f}(\cdot)$ is a continuously differentiable vector function of $\mathbf{x}$ and $\mathbf{u}$, and $\mathbf{h}(\cdot)$ is a continuously differentiable vector function of $\mathbf{x}$. It is assumed that the system (3) has an equilibrium point at origin, and the neighborhood of the equilibrium point is denoted as $\mathbb{B}(\mathbf{0}, \varepsilon)$. We also assume that the whole state $\mathbf{x}$ can be measured and is available for control. In the nonlinear regulation problem, we want to seek a feedback control $\mathbf{u} = \phi(\mathbf{x})$ to stabilize the origin, where $\phi(\cdot)$ is a function of $\mathbf{x}$. The Jacobin matrixes, which are assumed to be detectable, are denoted as

$$A(\mathbf{x}, \mathbf{u}) \triangleq \frac{\partial \mathbf{f}}{\partial \mathbf{x}}, B(\mathbf{x}, \mathbf{u}) \triangleq \frac{\partial \mathbf{f}}{\partial \mathbf{u}}, C(\mathbf{x}) \triangleq \frac{\partial \mathbf{h}}{\partial \mathbf{x}}, \Gamma(\mathbf{x}) \triangleq \frac{\partial \phi}{\partial \mathbf{x}}. \quad (4)$$

Assume that the elements in these matrixes are functions of $\mathbf{x}$, and continuous for all $\mathbf{x} \in \mathbb{R}^n$. Consider the Riccati equation as follow

$$\begin{aligned} P(\mathbf{x})A(\mathbf{x}, \mathbf{u}) + A(\mathbf{x}, \mathbf{u})^T P(\mathbf{x}) \\ - P(\mathbf{x})B(\mathbf{x}, \mathbf{u})R^{-1}B(\mathbf{x}, \mathbf{u})^T P(\mathbf{x}) + C(\mathbf{x})^T Q C(\mathbf{x}) = 0, \end{aligned} \quad (5)$$

where Q and R have the same define as equation (1).

Lemma 3 If $(A(\mathbf{x}, \mathbf{u}), B(\mathbf{x}, \mathbf{u}))$ is complete controllable, then there exists a solution $P(\mathbf{x}) \geq 0$ to (5).

Proof (Proof by Contradiction)

Assume to the contrary that there does not exist a solution $P(\mathbf{x}) \geq 0$ to (5) if $(A(\mathbf{x}, \mathbf{u}), B(\mathbf{x}, \mathbf{u}))$ is complete controllable. If this is the case, for any measured state $\hat{\mathbf{x}}$ of the nonlinear system, the pair $(\hat{A}, \hat{B})$ is controllable, and there have no solution $P \geq 0$ to the equation as follow

$$P\hat{A} + \hat{A}^T P - P\hat{B}R^{-1}\hat{B}^T P + \hat{C}^T Q \hat{C} = 0, \quad (6)$$

where

$$\hat{A} = A(\mathbf{x}, \mathbf{u})|_{\substack{\mathbf{x}=\hat{\mathbf{x}} \\ \mathbf{u}=\hat{\mathbf{u}}}}, \hat{B} = B(\mathbf{x}, \mathbf{u})|_{\substack{\mathbf{x}=\hat{\mathbf{x}} \\ \mathbf{u}=\hat{\mathbf{u}}}}, \hat{C} = C(\mathbf{x})|_{\mathbf{x}=\hat{\mathbf{x}}}, \quad (7)$$

and where $\hat{\mathbf{u}} = \phi(\hat{\mathbf{x}})$.

Note that $\hat{A}$ and $\hat{B}$ are constant matrix, and the solution to the equation (6) is a typical optimal LQR problem. Obviously, our hypothesis contradicts Lemma 1, so the original proposition is true.

This completes the proof.

Lemma 4 If $(A(\mathbf{x}, \mathbf{u}), B(\mathbf{x}, \mathbf{u}))$ is complete controllable and $(C(\mathbf{x}), A(\mathbf{x}, \mathbf{u}))$ is completely observable, then there exists a unique solution $P(\mathbf{x}) > 0$ to (5) and it yields an asymptotically stable closed-loop system.

Proof The proof can be completed according to Lemma 2 by using the method of proof by contradiction, which is similar to the proof of Lemma 3.

Theorem 1 If $(\mathbf{x}, \mathbf{u}) \in \mathbb{B}(\mathbf{0}, \varepsilon)$, the pair $(A(\mathbf{x}, \mathbf{u}), B(\mathbf{x}, \mathbf{u}))$ is controllable and the pair $(C(\mathbf{x}), A(\mathbf{x}, \mathbf{u}))$ is observable, then the origin of the nonlinear system (3) will be locally asymptotically stable; if for all $\mathbf{x}$ and $\mathbf{u}$ ($\mathbf{u} = \phi(\mathbf{x})$), the pair $(A(\mathbf{x}, \mathbf{u}), B(\mathbf{x}, \mathbf{u}))$ is controllable and the pair $(C(\mathbf{x}), A(\mathbf{x}, \mathbf{u}))$ is observable, then the origin of the nonlinear system (3) will be globally asymptotically stable.

Proof Assume that $P(\mathbf{x})$ is a symmetric positive definite matrix, the elements of $P(\mathbf{x})$ is function of $\mathbf{x}$, and $P(\mathbf{x})$ satisfies $\dot{p} = \frac{dP(\mathbf{x})}{dt} = \mathbf{0}$. We define the Lyapunov function V as follow

$$V(\mathbf{x}) = \mathbf{f}^T(\mathbf{x}, \mathbf{u})P(\mathbf{x})\mathbf{f}(\mathbf{x}, \mathbf{u}). \quad (8)$$

Since

$$\begin{aligned}\dot{\mathbf{f}}(\mathbf{x}, \mathbf{u}) &= \frac{d\mathbf{f}(\mathbf{x}, \mathbf{u})}{dt} \\ &= \frac{\partial \mathbf{f}(\mathbf{x}, \mathbf{u})}{\partial \mathbf{x}} \frac{d\mathbf{x}}{dt} + \frac{\partial \mathbf{f}(\mathbf{x}, \mathbf{u})}{\partial \mathbf{u}} \frac{\partial \mathbf{u}}{\partial \mathbf{x}} \frac{d\mathbf{x}}{dt} \\ &= [A(\mathbf{x}, \mathbf{u}) + B(\mathbf{x}, \mathbf{u})\Gamma(\mathbf{x})]\dot{\mathbf{x}} \\ &= [A(\mathbf{x}, \mathbf{u}) + B(\mathbf{x}, \mathbf{u})\Gamma(\mathbf{x})]\mathbf{f},\end{aligned}\quad (9)$$

We obtain

$$\begin{aligned}\dot{V} &= \frac{dV(\mathbf{x})}{dt} \\ &= \mathbf{f}(\mathbf{x}, \mathbf{u})^T P(\mathbf{x}) \dot{\mathbf{f}}(\mathbf{x}, \mathbf{u}) + \dot{\mathbf{f}}(\mathbf{x}, \mathbf{u})^T P(\mathbf{x}) \mathbf{f}(\mathbf{x}, \mathbf{u}) \\ &\quad + \mathbf{f}(\mathbf{x}, \mathbf{u})^T \dot{P}(\mathbf{x}) \mathbf{f}(\mathbf{x}, \mathbf{u}) \\ &= \mathbf{f}(\mathbf{x}, \mathbf{u})^T P(\mathbf{x}) [A(\mathbf{x}, \mathbf{u}) + B(\mathbf{x}, \mathbf{u})\Gamma(\mathbf{x})] \mathbf{f}(\mathbf{x}, \mathbf{u}) \\ &\quad + \mathbf{f}(\mathbf{x}, \mathbf{u})^T [A(\mathbf{x}, \mathbf{u})^T + \Gamma(\mathbf{x})^T B(\mathbf{x}, \mathbf{u})^T] P(\mathbf{x}) \mathbf{f}(\mathbf{x}, \mathbf{u}) \\ &= \mathbf{f}(\mathbf{x}, \mathbf{u})^T [P(\mathbf{x})A(\mathbf{x}, \mathbf{u}) + P(\mathbf{x})B(\mathbf{x}, \mathbf{u})\Gamma(\mathbf{x}) \\ &\quad + A(\mathbf{x}, \mathbf{u})^T P(\mathbf{x}) + \Gamma(\mathbf{x})^T B(\mathbf{x}, \mathbf{u})^T P(\mathbf{x})] \mathbf{f}(\mathbf{x}, \mathbf{u}).\end{aligned}\quad (10)$$

Let

$$\begin{aligned}J &= P(\mathbf{x})A(\mathbf{x}, \mathbf{u}) + P(\mathbf{x})B(\mathbf{x}, \mathbf{u})\Gamma(\mathbf{x}) \\ &\quad + A(\mathbf{x}, \mathbf{u})^T P(\mathbf{x}) + \Gamma(\mathbf{x})^T B(\mathbf{x}, \mathbf{u})^T P(\mathbf{x}).\end{aligned}$$

If J is a negative semidefinite matrix (ie. $J \leq 0$), we obtain $\dot{V} = \mathbf{f}(\mathbf{x}, \mathbf{u})^T J \mathbf{f}(\mathbf{x}, \mathbf{u}) \leq 0$. Thus, we can conclude that the origin of the nonlinear system (3) will be asymptotically stable. Therefore, the core of the subject is if there exists a positive definite solution $P(\mathbf{x}) > 0$, such that

$$\begin{aligned}J &= P(\mathbf{x})A(\mathbf{x}, \mathbf{u}) + P(\mathbf{x})B(\mathbf{x}, \mathbf{u})\Gamma(\mathbf{x}) \\ &\quad + A(\mathbf{x}, \mathbf{u})^T P(\mathbf{x}) + \Gamma(\mathbf{x})^T B(\mathbf{x}, \mathbf{u})^T P(\mathbf{x}) \leq 0.\end{aligned}\quad (11)$$

Since $\mathbf{u}$ is unconstrained control input of the system, the $\Gamma(\mathbf{x})$ can be set as

$$\Gamma(\mathbf{x}) = -K(\mathbf{x}), \quad (12)$$

where $K(\mathbf{x})$ is given by

$$K(\mathbf{x}) = R^{-1}B(\mathbf{x}, \mathbf{u})^T P(\mathbf{x}), \quad (13)$$

and R is an $m \times m$ symmetric positive definite matrix. Substituting (12) through (13) into (11) yields

$$\begin{aligned}P(\mathbf{x})A(\mathbf{x}, \mathbf{u}) + A(\mathbf{x}, \mathbf{u})^T P(\mathbf{x}) \\ - 2P(\mathbf{x})B(\mathbf{x}, \mathbf{u})R^{-1}B(\mathbf{x}, \mathbf{u})^T P(\mathbf{x}) \leq 0.\end{aligned}\quad (14)$$

Consider the following equation

$$\begin{aligned}P(\mathbf{x})A(\mathbf{x}, \mathbf{u}) + A(\mathbf{x}, \mathbf{u})^T P(\mathbf{x}) \\ - 2P(\mathbf{x})B(\mathbf{x}, \mathbf{u})R^{-1}B(\mathbf{x}, \mathbf{u})^T P(\mathbf{x}) = -C(\mathbf{x})^T Q C(\mathbf{x}),\end{aligned}\quad (15)$$

where Q is an $l \times l$ symmetric positive definite matrix. We can rewrite (15) as

$$\begin{aligned}P(\mathbf{x})A(\mathbf{x}, \mathbf{u}) + A(\mathbf{x}, \mathbf{u})^T P(\mathbf{x}) \\ - P(\mathbf{x})B(\mathbf{x}, \mathbf{u})(\frac{1}{2}R)^{-1}B(\mathbf{x}, \mathbf{u})^T P(\mathbf{x}) + C(\mathbf{x})^T Q C(\mathbf{x}) = 0.\end{aligned}\quad (16)$$

Note that $C(\mathbf{x})^T Q C(\mathbf{x})$ is a semidefinite matrix. Since the pair $(A(\mathbf{x}, \mathbf{u}), B(\mathbf{x}, \mathbf{u}))$ is controllable and the pair $(C(\mathbf{x}), A(\mathbf{x}, \mathbf{u}))$ is observable, there exists a unique positive definite solution $P(\mathbf{x})$ to the equation (16) according to Lemma 4, and the inequality (14) will be satisfied. Hence, the inequality (11) will be satisfied in turn.

This completes the proof.

At the origin (i.e., equilibrium point), system (3) can be linearized as

$$\begin{aligned}\dot{\mathbf{x}} &= A^0 \mathbf{x} + B^0 \mathbf{u}, \\ \mathbf{y} &= C^0 \mathbf{x},\end{aligned}\quad (17)$$

where $A^0 = A(\mathbf{x}, \mathbf{u})|_{\mathbf{x}=0, \mathbf{u}=0}$, $B^0 = B(\mathbf{x}, \mathbf{u})|_{\mathbf{x}=0, \mathbf{u}=0}$, $C^0 = C(\mathbf{x})|_{\mathbf{x}=0}$.

Theorem 2 Assume that the pair (A^0, B^0) is controllable. There exists some $\varepsilon > 0$ such that for any $(\mathbf{x}, \mathbf{u}) \in \mathbb{B}(\mathbf{0}, \varepsilon)$, the pair $(A(\mathbf{x}, \mathbf{u}), B(\mathbf{x}, \mathbf{u}))$ is also controllable.

Proof Take the controllability matrix

$$W_{\text{ctrl}}(\mathbf{x}, \mathbf{u}) = \begin{bmatrix} B(\mathbf{x}, \mathbf{u}) & A(\mathbf{x}, \mathbf{u})B(\mathbf{x}, \mathbf{u}) \\ \vdots & A(\mathbf{x}, \mathbf{u})^{n-1}B(\mathbf{x}, \mathbf{u}) \end{bmatrix}. \quad (18)$$

Since $\mathbf{f}(\cdot)$ is continuous, the elements of matrix $W_{\text{ctrl}}(\mathbf{x}, \mathbf{u})$ are continuous functions of $\mathbf{x}$ and $\mathbf{u}$. In addition, since (A^0, B^0) is controllable, the controllability matrix W_{ctrl} is of full row rank, i.e.,

$$\text{rank} \left(W_{\text{ctrl}}(\mathbf{x}, \mathbf{u})|_{\mathbf{x}=0, \mathbf{u}=0} \right) = n.$$

Hence, a $n \times n$ submatrix $W_{\text{ctrl}}^*(\mathbf{x}, \mathbf{u})$, which satisfies $\det \left(W_{\text{ctrl}}^*(\mathbf{x}, \mathbf{u})|_{\mathbf{x}=0, \mathbf{u}=0} \right) \neq 0$, can be found in $W_{\text{ctrl}}(\mathbf{x}, \mathbf{u})$. Let

$$W_{\text{ctrl}}^0 \triangleq W_{\text{ctrl}}^*(\mathbf{x}, \mathbf{u})|_{\mathbf{x}=0, \mathbf{u}=0}, \quad \omega(\mathbf{x}, \mathbf{u}) \triangleq \det(W_{\text{ctrl}}^*(\mathbf{x}, \mathbf{u})). \quad (19)$$

Since $\det(W_{\text{ctrl}}^0) \neq 0$, we have $\omega(\mathbf{x}, \mathbf{u})|_{\mathbf{x}=0, \mathbf{u}=0} \neq 0$. Because $\omega(\mathbf{x}, \mathbf{u})$ is continuous function of $\mathbf{x}$ and $\mathbf{u}$, there exists $\varepsilon > 0$ such that for any $(\mathbf{x}, \mathbf{u}) \in \mathbb{B}(\mathbf{0}, \varepsilon)$, $\omega(\mathbf{x}, \mathbf{u}) \neq 0$.

Therefore, for any $(\mathbf{x}, \mathbf{u}) \in \mathbb{B}(\mathbf{0}, \varepsilon)$, $\text{rank}(W_{\text{ctrl}}(\mathbf{x}, \mathbf{u})) = n$, i.e., the pair $(A(\mathbf{x}, \mathbf{u}), B(\mathbf{x}, \mathbf{u}))$ is controllable.

This completes the proof.

Theorem 3 Assume that the pair (C^0, A^0) is observable. There exists some $\varepsilon > 0$ such that for any $(\mathbf{x}, \mathbf{u}) \in \mathbb{B}(\mathbf{0}, \varepsilon)$, the pair $(C(\mathbf{x}), A(\mathbf{x}, \mathbf{u}))$ is also observable.

Proof Consider the observability matrix

$$W_{\text{obsv}}(\mathbf{x}, \mathbf{u}) = [C(\mathbf{x}) \ C(\mathbf{x})A(\mathbf{x}, \mathbf{u}) \ \cdots \ C(\mathbf{x})A(\mathbf{x}, \mathbf{u})^{n-1}]^T.$$

The rest proof is similar to that of Theorem 2.

Theorem 4 With regard to nonlinear (3), if the pair $(A(\mathbf{x}, \mathbf{u}), B(\mathbf{x}, \mathbf{u}))$ is controllable and the pair $(C(\mathbf{x}), A(\mathbf{x}, \mathbf{u}))$ is observable, and assumed that the initial state is $(\mathbf{x}(0) = \mathbf{0}, \mathbf{u}(0) = \mathbf{0})$, then a stabilizing state feedback control law for the nonlinear system is

$$\mathbf{u} = -K(\mathbf{x})\mathbf{x}, \quad (20)$$

where $K(\mathbf{x}) = R^{-1}B(\mathbf{x}, \mathbf{u})^T P(\mathbf{x})$, and $P(\mathbf{x})$ is the unique positive definite solution to the following equation

$$\begin{aligned}P(\mathbf{x})A(\mathbf{x}, \mathbf{u}) + A(\mathbf{x}, \mathbf{u})^T P(\mathbf{x}) \\ - P(\mathbf{x})B(\mathbf{x}, \mathbf{u})R^{-1}B(\mathbf{x}, \mathbf{u})^T P(\mathbf{x}) + C(\mathbf{x})^T Q C(\mathbf{x}) = 0,\end{aligned}\quad (21)$$

where Q and R are selected positive definite matrixes.

Proof According to Lemma 4, we know there exists a unique solution $P > 0$ to (21). Let

$$P(\mathbf{x})B(\mathbf{x}, \mathbf{u})R^{-1}B(\mathbf{x}, \mathbf{u})^T P(\mathbf{x}) + C(\mathbf{x})^T Q C(\mathbf{x}) = M.$$

Obviously, M is a positive semidefinite matrix, i.e., $M \geq 0$. Then, the equation (21) can be written as

$$\begin{aligned}P(\mathbf{x})A(\mathbf{x}, \mathbf{u}) + A(\mathbf{x}, \mathbf{u})^T P(\mathbf{x}) \\ - 2P(\mathbf{x})B(\mathbf{x}, \mathbf{u})R^{-1}B(\mathbf{x}, \mathbf{u})^T P(\mathbf{x}) = -M.\end{aligned}\quad (22)$$

Thus, the solution $P > 0$ to equation (21) can ensure

$$\begin{aligned}P(\mathbf{x})A(\mathbf{x}, \mathbf{u}) + A(\mathbf{x}, \mathbf{u})^T P(\mathbf{x}) \\ - 2P(\mathbf{x})B(\mathbf{x}, \mathbf{u})R^{-1}B(\mathbf{x}, \mathbf{u})^T P(\mathbf{x}) \leq 0.\end{aligned}\quad (23)$$

Note that the form of equation(23) is the same as that of equation (14). Take the same Lyapunov function V as that of Theorem 1. In this case, the derivative $\dot{V}$ satisfies $\dot{V} \leq 0$. Then, according to equation (12) and (13), we can obtain a stabilizing state feedback control law

$$\mathbf{u}(t) = \int_{\mathbf{x}(t_0)}^{\mathbf{x}(t)} -K(\xi)d\xi, \quad \mathbf{x}(t_0) = \mathbf{0}, \mathbf{u}(t_0) = \mathbf{0}, \quad (24)$$

where $K(\mathbf{x})$ is taken as (??), which can be rewritten as

$$K(\mathbf{x}(t)) = R^{-1}B(\mathbf{x}(t), \phi(\mathbf{x}(t)))^T P(\mathbf{x}(t)). \quad (25)$$

In practice all measurements are only taken at discrete sampling instants, and only the numerical solution of $P(\mathbf{x}(t_i))$ can be obtained in every sampling interval. Therefore, the control action must be piecewise constant.

Spouse that the time dimension $[t_0, t]$ is divided into

$$0 = t_0 < t_1 < \dots < t_n = t,$$

and the sampling interval is $\tau_i \triangleq t_i - t_{i-1}$. Thus, at every sampling time t_i the state $\mathbf{x}(t_i)$ and $K(\mathbf{x}(t_i))$ can be obtained as

$$\begin{aligned} \mathbf{x}(t_0) = \mathbf{0}, \quad \mathbf{x}(t_1), \quad \dots, \quad \mathbf{x}(t_n) = \mathbf{x}(t), \\ K(\mathbf{x}(t_0)) = \mathbf{0}, \quad K(\mathbf{x}(t_1)), \quad \dots, \quad K(\mathbf{x}(t_n)) = K(\mathbf{x}(t)). \end{aligned}$$

Note that the sequence of $\mathbf{x}(t_i)$ and $K(\mathbf{x}(t_i))$ don't have monotonicity. Since $K(\mathbf{x})$ is continuous for $\mathbf{x}$, we have

$$\begin{aligned} \mathbf{u}(t) &= \int_{\mathbf{x}(t_0)}^{\mathbf{x}(t)} -K(\xi)d\xi \\ &= - \left(\int_{\mathbf{x}(t_0)}^{\mathbf{x}(t_1)} K(\xi)d\xi + \int_{\mathbf{x}(t_1)}^{\mathbf{x}(t_2)} K(\xi)d\xi + \dots + \int_{\mathbf{x}(t_{n-1})}^{\mathbf{x}(t_n)} K(\xi)d\xi \right). \end{aligned}$$

Applying the mean value theorem, yields

$$\begin{aligned} \mathbf{u}(t) &= -(K(\eta_1)(\mathbf{x}_1 - \mathbf{x}_0) + K(\eta_2)(\mathbf{x}_2 - \mathbf{x}_1) \\ &\quad + \dots + K(\eta_n)(\mathbf{x}_n - \mathbf{x}_{n-1})), \end{aligned} \quad (26)$$

where $\mathbf{x}(t_i)$ is denoted as $\mathbf{x}_i$ and η_i is between $\mathbf{x}_i$ and $\mathbf{x}_{i-1}$. When $\max(\tau_i)$ is sufficient small, the $\Delta \mathbf{x}_i \triangleq \mathbf{x}_i - \mathbf{x}_{i-1}$ is sufficient small, and $\Delta K(\mathbf{x}_i) \triangleq K(\mathbf{x}_i) - K(\mathbf{x}_{i-1})$ is also sufficient small. Therefore, when n tends to the infinity (i.e., $n \rightarrow \infty$), and the $\max(\tau_i)$ is sufficient small, we have

$$\begin{aligned} \eta_i &\rightarrow \mathbf{x}_i, & \text{and} & \quad \eta_i \rightarrow \mathbf{x}_{i-1}, \\ K(\eta_i) &\rightarrow K(\mathbf{x}_i), & \text{and} & \quad K(\eta_i) \rightarrow K(\mathbf{x}_{i-1}), \end{aligned}$$

then yields

$$K(\eta_i)(\mathbf{x}_i - \mathbf{x}_{i-1}) \approx K(\eta_i)\mathbf{x}_1 - K(\eta_{i-1})\mathbf{x}_{i-1}. \quad (27)$$

Substituting (27) into (26) yields

$$\begin{aligned} \mathbf{u}(t) &\approx -([K(\mathbf{x}_1)\mathbf{x}_1 - K(\mathbf{x}_0)\mathbf{x}_0] + [K(\mathbf{x}_2)\mathbf{x}_2 - K(\mathbf{x}_1)\mathbf{x}_1] \\ &\quad + \dots + [K(\mathbf{x}_n)\mathbf{x}_n - K(\mathbf{x}_{n-1})\mathbf{x}_{n-1}]) \\ &\approx -(K(\mathbf{x}_n)\mathbf{x}_n - K(\mathbf{x}_0)\mathbf{x}_0) \\ &\approx -K(\mathbf{x}(t_n))\mathbf{x}(t_n). \end{aligned}$$

Therefore, when $\max(\tau_i) \rightarrow 0$, we have

$$\mathbf{u}(t) = \int_{\mathbf{x}(t_0)}^{\mathbf{x}(t)} -K(\xi)d\xi = -K(\mathbf{x}(t))\mathbf{x}(t). \quad (28)$$

This completes the proof.

The VGLQR method is summarized as follow:

- i) Derive the Jacobin matrixes $A(\mathbf{x}, \mathbf{u})$, $B(\mathbf{x}, \mathbf{u})$ and $C(\mathbf{x})$ from the the model of the nonlinear system.
- ii) Solve the Riccati equation (21) to obtain $P(\mathbf{x}) > 0$.
- iii) Construct the nonlinear feedback controller via

$$\mathbf{u} = -R^{-1}B(\mathbf{x}, \mathbf{u})^T P(\mathbf{x})(\mathbf{x}). \quad (29)$$

Remark The above steps are finished online.

III. THE ALGORITHM FOR THE SOLUTION OF ARE

The problem for the VGLQR method will come down to solving the algebraic Riccati equation. However, with respect to a large scale algebraic Riccati equation, there are many problems, such as operation speed, precision, storage space, the convergence rate and the initial guess of the closed loop system gain, etc. Therefore, it is very difficult to solve the algebraic equation on line. The following sections will review two widely used numerical techniques for solving Riccati equation and a fast algorithm based on Schur method is designed.

A. Kleinman iterative algorithm

The Kleinman iterative algorithm is described by Kleinman [21], and only the features are listed below for being compared.

As to Kleinman iterative algorithm, the only positive definite solution for algebraic Riccati equations can be guaranteed if a proper stable initial gain matrix is selected for the closed loop system. The essential of the algorithm is using the Newton iteration method to solve the algebraic Riccati equation, and the monotonic convergence of the iterative sequence with a faster speed is guaranteed. Depending upon the good initial guess, 10 or more iterations may be required to obtain the solution. Some available methods, such as pole placement and Schur algorithm etc., are often used to obtain the initial gain, and a better quality initial guess can also produce faster convergence of the algorithm. Therefore, the main disadvantage of the algorithm is to determine the initial gain matrix, and this will increase the additional operation counts and lead to a more complicated solution.

B. Schur method and the fast algorithm

Schur method [22] is a direct method, which is especially suitable for higher order systems. The algorithm is accurate, stable, and unnecessary for a good quality initial gain matrix. Whether the open loop system is stable or not, the positive definite solution for algebraic Riccati equation can be obtained. But the computation and storage requirement for this algorithm can be more than twice as much as that of Kleinman iterative algorithm. It is estimated that floating-point operations are required for a high dimensional system, which is the main barrier for its computation on line. Considering the obvious advantages and overcoming the disadvantages for implement, a fast algorithm of Schur method for the solution of algebraic Riccati equations, which is suitable for computation on line, is designed in this paper and successfully implemented in the triple inverted pendulum control system. In this paper, only the essential principle of the fast algorithm based on Schur method is briefly introduced below.

Consider the algebraic Riccati equation

$$PA + A^T P - PBR^{-1}B^T P + C^T QC = 0. \quad (30)$$

The Hamiltonian matrix corresponding to (30) is defined as

$$Z = \begin{bmatrix} A & -BR^{-1}B^T \\ -C^T QC & -A^T \end{bmatrix}_{2n \times 2n}. \quad (31)$$

Assume that Z has no pure imaginary eigenvalues. Then, an orthogonal transformation U can be found to put Z in real Schur form

$$U^T Z U = \begin{bmatrix} S_{11} & S_{12} \\ \mathbf{0} & S_{22} \end{bmatrix}_{2n \times 2n}, \quad (32)$$

where S_{ij} is a $n \times n$ matrix. It is possible to arrange that the real parts of the spectrum of S_{11} are negative while the real parts

of the spectrum of S_{22} are positive. Moreover, U is conformably partitioned into four $n \times n$ blocks

$$U = \begin{bmatrix} U_{11} & U_{12} \\ U_{21} & U_{22} \end{bmatrix}_{2n \times 2n}. \quad (33)$$

Then, the eigenvalues of S_{11} are the stable poles of the closed system, and the solution for algebraic Riccati equation (30) is

$$P = U_{21}U_{11}^{-1}. \quad (34)$$

According to the above discussion, the steps of the algorithm for solving algebraic Riccati equations are as follows:

Step 1 Construct the Hamiltonian matrix Z ;

Step 2 Find an orthogonal transformation U_1 which reduce Z to an upper Hessenberg matrix S_1 , i.e., $U_1^T Z U_1 = S_1$;

Step 3 Find an orthogonal transformation U_2 which reduce S_1 to real Schur form S_2 by using the double Francis QR algorithm, i.e., $U_2^T S_1 U_2 = S_2$;

Step 4 Find an orthogonal transformation U_3 which reduce S_2 to strictly upper triangular matrix S_3 by using Givens transformation, i.e., $U_3^T S_2 U_3 = S_3$;

Step 5 Find an orthogonal transformation U_4 which arrange S_3 in strictly upper triangular form S , where the real parts of the spectrum of S_{11} are negative and the real parts of the spectrum of S_{22} are positive, i.e., $U_4^T S_3 U_4 = S$;

Step 6 Let $U = U_1 U_2 U_3 U_4$. Then $U^T Z U = S$. If U is a unitary matrix, the real part of the matrix should be separated at first and then P is derived according to (34).

IV. STABILIZATION FOR TRIPLE INVERTED PENDULUM

A triple inverted pendulum, a highly sensitive system, is stabilized by using the VGLQR method, which demonstrates the potential of the control scheme.

As shown in Figure 3, the triple inverted pendulum is consisted of a cart, rod₁, rod₂ and rod₃, which are linked by hinges. The cart can move freely on the linear orbit, and the rods can rotate freely in plumb plane. Let the clockwise angle and moment in Figure 3 be in positive direction. And let x be the displacement of the cart, u the acceleration of the cart, θ_i the angle between rod _{i} and vertical direction, O_i and G_i the linking point and centroid of rod _{i} respectively, m_i the mass of rod _{i} , J_i the moment of inertia of rod _{i} around G_i , l_i the distance from O_i to G_i , L_i the length of rod _{i} , c_i the friction coefficient of rod _{i} around O_i , ($i = 1, 2, 3$). The equilibrium point of the triple inverted pendulum is that the rod₁, rod₂ and rod₃ are all in upright position and the cart's position is designated as zero point. The mechanical parameters are described in Table 1.

The model of the triple pendulum is represented as follow

$$\ddot{x} = u, \quad (35)$$

$$b_1 \ddot{\theta}_1 + a_2 L_1 \cos(\theta_2 - \theta_1) \ddot{\theta}_2 + a_3 L_1 \cos(\theta_3 - \theta_1) \ddot{\theta}_3 = (-c_1 - c_2) \dot{\theta}_1 + a_2 L_1 \sin(\theta_2 - \theta_1) \dot{\theta}_2^2 + c_2 \dot{\theta}_2 + a_3 L_1 \sin(\theta_3 - \theta_1) \dot{\theta}_3^2 + a_1 g \sin \theta_1 - a_1 \cos \theta_1 \ddot{x}, \quad (36)$$

$$a_3 L_1 \cos(\theta_3 - \theta_1) \ddot{\theta}_1 + a_3 L_2 \cos(\theta_3 - \theta_2) \ddot{\theta}_2 + b_3 \ddot{\theta}_3 = -a_3 L_1 \sin(\theta_3 - \theta_1) \dot{\theta}_1^2 - a_3 L_2 \sin(\theta_3 - \theta_2) \dot{\theta}_2^2 + c_3 \dot{\theta}_2 - c_3 \dot{\theta}_3 + a_3 g \sin \theta_3 - a_3 \cos \theta_3 \ddot{x}, \quad (37)$$

$$a_3 L_1 \cos(\theta_3 - \theta_1) \ddot{\theta}_1 + a_3 L_2 \cos(\theta_3 - \theta_2) \ddot{\theta}_2 + b_3 \ddot{\theta}_3 = -a_3 L_1 \sin(\theta_3 - \theta_1) \dot{\theta}_1^2 - a_3 L_2 \sin(\theta_3 - \theta_2) \dot{\theta}_2^2 + c_3 \dot{\theta}_2 - c_3 \dot{\theta}_3 + a_3 g \sin \theta_3 - a_3 \cos \theta_3 \ddot{x}, \quad (38)$$

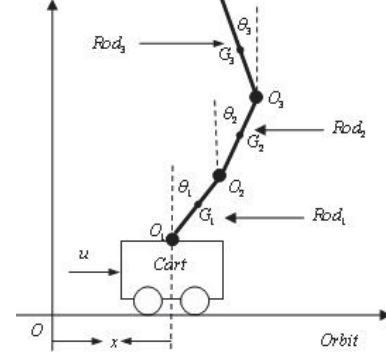


Fig. 1. The simplified structure of triple inverted pendulum

TABLE I
MECHANICAL PARAMETERS OF THE TRIPLE PENDULUM

	rod ₁	rod ₂	rod ₃
L_i (m)	0.1786	0.2706	0.6463
l_i (m)	0.150	0.213	0.300
m_i (kg)	0.27340	0.31068	0.149715
J_i (Nms ²)	0.0022474	0.0041544	0.0056056
c_i (Nms)	0.002646	0.002646	0.002646

where

$$a_i \triangleq m_i l_i + \sum_{j=i+1}^3 m_j L_j, \quad 1 \leq i \leq 3,$$

$$b_i \triangleq J_i + m_i l_i^2 + \sum_{j=i+1}^3 m_j L_j^2, \quad 1 \leq i \leq 3.$$

Taking $\mathbf{x} = [x, \theta_1, \theta_2, \theta_3, \dot{x}, \dot{\theta}_1, \dot{\theta}_2, \dot{\theta}_3]^T$, the (35)-(38) can be rewritten as the state-space form

$$\begin{aligned} \dot{\mathbf{x}} &= \mathbf{f}(\mathbf{x}, u), \\ \mathbf{y} &= \mathbf{x}. \end{aligned} \quad (39)$$

where $\mathbf{f} = [f_1, f_2, \dots, f_8]^T$.

In the numerical experiment, we take $Q = \text{diag}(10, 500, 500, 500, 0, 0, 0, 0)$, $R = 1$, and apply the VGLQR and LQR method to stabilize the triple inverted pendulum, and the initial state is taken as: $x = 0$, $\theta_1 = \theta_3 = 0.0536(\text{rad})$ and $\theta_2 = -0.0536(\text{rad})$, which is a difficult condition for control. The simulation results are shown in Figure 2(a)-(e). As can be seen from Figure 2(a)-(e), the VGLQR controller has better adaptiveness and robustness.

In addition, The effectiveness of the VGLQR method has also been demonstrated by physical experiment. A triple inverted pendulum is successfully stabilized by using the present control scheme. The real-time responses of one physical experiment are shown in Figure 2(f)-(j).

Remark Because the friction and other features of our nonlinear simulation model can approximate but are unable to duplicate exactly the physical system's behaviour, the real-time curves fluctuate in a small range and can not converge to zero (see Figure 2(f)-(j)). In practice, since the uncertain factors always exert on the system during the real-time control process, stabilizing the three pendulums around the unstable equilibrium state is a dynamic process, and the cart and rods fluctuate near the equilibrium point.

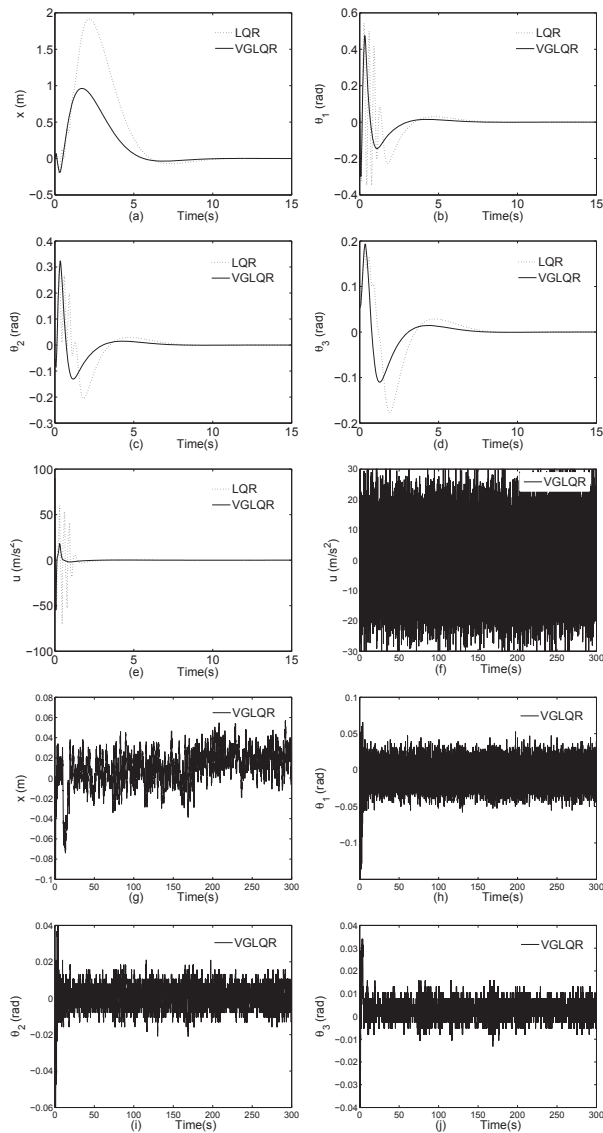


Fig. 2. Response curves of x , θ_1 , θ_2 , θ_3 , and u

V. CONCLUSIONS

For a class of nonlinear control systems, a variable gain linear quadratic regulator(VGLQR) control method is proposed to obtain more precise dynamic feedback control. Moreover, a fast algorithm based on Schur method for real-time solving algebraic Riccati equation is discussed and the program is coded by C++. Three application examples are used to demonstrate the potentiality of the proposed control method. Especially in stabilizing of a triple inverted pendulum, the VGLQR method displays a better regulation capacity for a highly sensitive system.

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